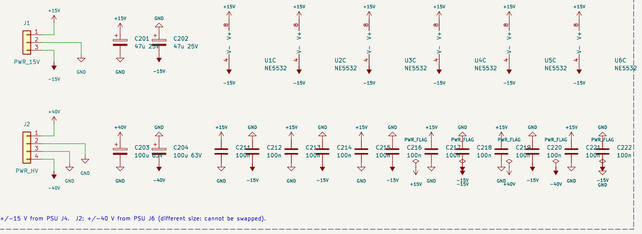
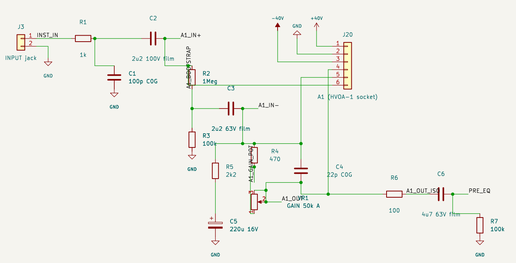
Power input and op-amp decoupling



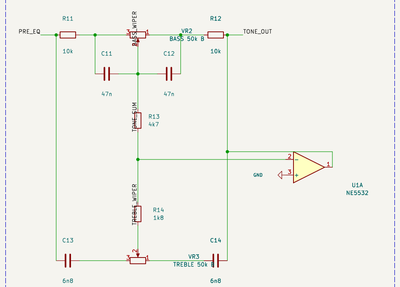
J1: +/-15 V from PSU J4. J2: +/-40 V from PSU J6 (different size; cannot be swapped).

A1: bootstrapped instrument input / gain (+1.7 .. +27 dB)



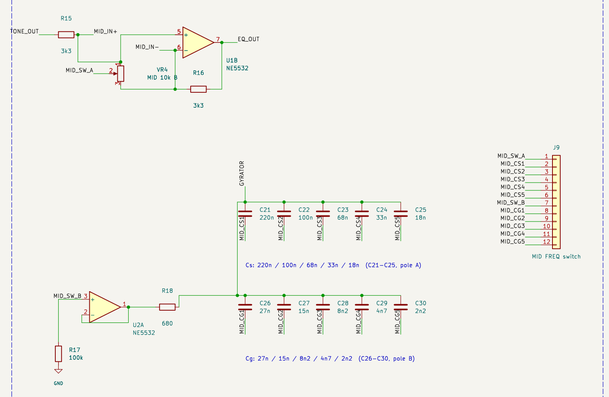
RA (R2) is bootstrapped by C3 from IN− : >1 MOhm input impedance.
VR1 is a rheostat: gain rises CW. C2/C3: film, 100 V.

EQ: Baxandall bass / treble (U1A)



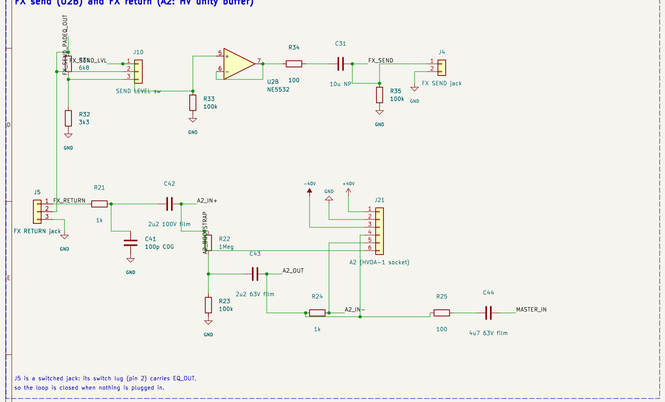
CW = boost (wiper toward the input end).

EQ: mid, gyrator with 5-way frequency switch (U1B, U2A)



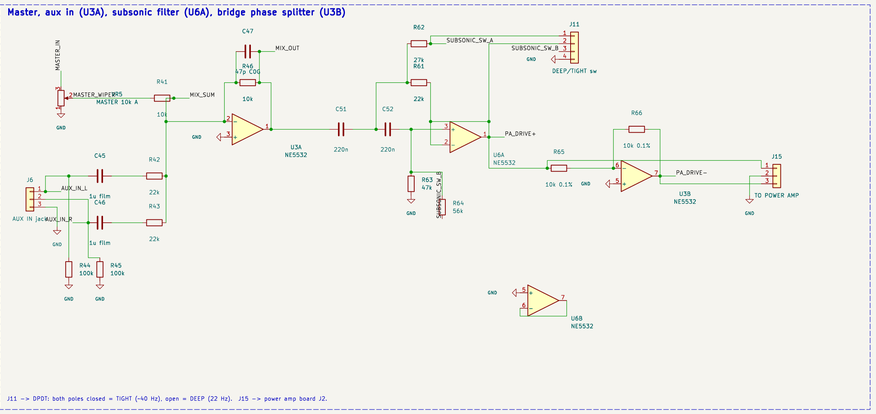
J9 -> 2-pole rotary: pole A (MW) selects Cs, pole B (GP) selects Cg.
250 / 500 / 800 / 1.5k / 3k Hz. Keep J9 wiring short and shielded.

FX send (U2B) and FX return (A2: HV unity buffer)



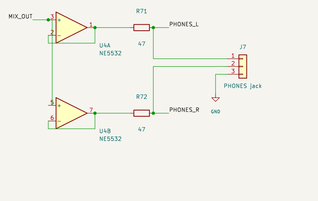
J5 is a switched jack: its switch lug (pin 2) carries EQ_OUT,
so the loop is closed when nothing is plugged in.

Master, aux in (U3A), subsonic filter (U6A), bridge phase splitter (U3B)



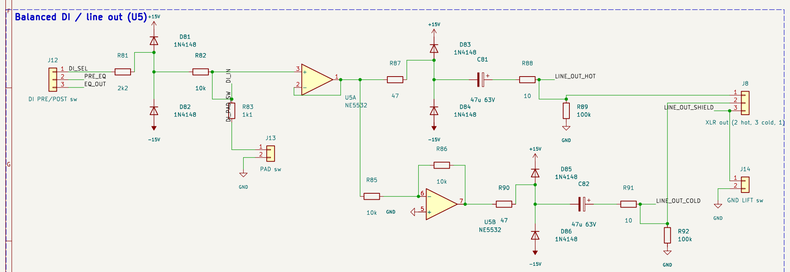
J11 -> DPDT: both poles closed = TIGHT (-40 Hz), open = DEEP (22 Hz). J15 -> power amp board J2.

Headphone amplifier (U4)



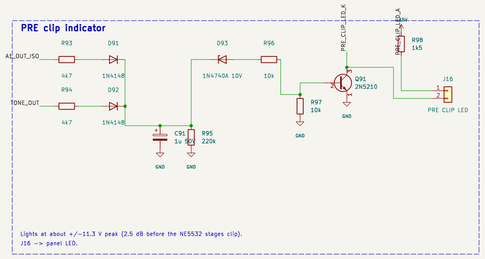
Phones jack switch contacts are wired to
the power amp board J5 (mutes the speaker).

Balanced DI / line out (U5)



J12: PRE/POST select. J13 closed = −20 dB pad. J14: ground-lift switch.
C81/C82 + toward the XLR (phantom power).

PRE clip indicator



Lights at about +/−11.3 V peak (2.5 dB before the NE5532 stages clip).
J16 -> panel LED.

NP = bipolar electrolytic (CP footprint, either orientation).
Keep the +/−40 V area away from the NE5532 area: every HV output enters the 15 V domain AC-coupled.
A1/A2 are HVOA-1 daughterboards (hvoa_module project) plugged into J20/J21.
Pots are PCB-mounted Alpha 16 mm (pin 3 = CW end). Jacks and switches are panel-mounted on J5.

Sheet: /	
File: preamp.kicad_sch	
Title: Bass amp preamp board (HV front end, EQ, I/O)	
Size: A1	Date: **Rev: 1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Id: 1/1